

identified in the roadmap for the development of green hydrogen technology. Specific measures necessary for the development of the entire green hydrogen value chain are also identified.

iii. Where applicable, national objectives with regard to competitiveness

Balancing the EU's climate ambition with industrial competitiveness is essential. The abundance and competitive prices of clean electricity for hydrogen production are an essential prerequisite for green ammonia to become competitive and to change current production technology. To realise the transition to green ammonia, broad support for investment and operational expenditure is needed. Of course, it is also important to create markets that stimulate production. But above all, there must be fruitful cooperation between the EU, Member States and regional authorities to support industries to introduce new technologies, such as providing financial incentives and cooperation in the development of hydrogen resources. Hydrogen produced from renewable sources is key to replacing natural gas. There is scope for Bulgaria to be among the first in Europe to build new installations and hydrogen hubs, also linked to the creation of new jobs with higher labour added-value. Hydrogen is most often obtained from a steam reforming methane (SMR), a process in which carbon dioxide is emitted. Green ammonia is produced from hydrogen, which is produced by hydro electrolysis powered by electricity from RES. To date, potential consumers of green hydrogen in Bulgaria to replace the SMR process they use are LUKOYL Neftochim Burgas, the largest refinery in south-eastern Europe and the azo-torso industry (Agropolychim AD, Neochim AD), which accounts for 3 % of European production.

3. POLICIES AND MEASURES

3.1. Decarbonisation dimension

3.1.1. GHG emissions and removals

i. Policies and measures to meet the targets under Regulation (EU) 2018/842 referred to in point 2.1.1 and policies and measures to comply with Regulation (EU) 2018/841, covering all major sources and sectors for the enhancement of removals, with an idea of the long-term vision and goal of a low-emission economy and striking a balance between emissions and removals in line with the Paris Agreement

Regulation (EU) 2018/842 on binding annual greenhouse gas emission reductions by Member States from 2021 to 2030 provides for a national reduction target of 0 % compared

to 2005. The adopted changes to Regulation (EU) 2023/857 make the target for¹ Bulgaria more ambitious and much more challenging, increasing it to – 10 %. To achieve the target, Bulgaria will use several measures already in place, such as new and additional policies, which will be put in place over time and which will require major efforts by the country.

The sectors responsible for greenhouse gas emissions covered by Regulation (EU) 2023/857 are: transport, buildings, non-ETS industry, waste, agriculture. Emissions from road and maritime transport as well as heating of buildings will be regulated in the context of the revision of the ETS Directive². The national policies and measures set out for these sectors will therefore complement the cap and emissions trading mechanism provided for in the new ETS Directive.

Follow the main sectoral policies and measures to achieve the 2030 greenhouse gas reduction target.

Transport Sector

Bulgaria's main policy objectives for reducing greenhouse gas emissions in the transport sector are:

- Promoting the production of electric and other environmentally friendly vehicles;
- Encouraging the use/demand of new environmentally friendly vehicles;
- Accelerated deployment of charging infrastructure for electric and hybrid cars;
- Promoting research and development activities related to environmentally friendly vehicles;
- Promoting the production of sustainable aviation fuels (SAF) by introducing mandatory blending mandates under Annex I of Regulation (EU) 2023/2405 of the EP and of the Council of 18 October 2023 on ensuring a level playing field for sustainable air transport (ReFuelEU Aviation);
- Promoting the supply and use of sustainable aviation fuels;
- Provision of refuelling infrastructure for sustainable aviation fuels at airports for public use;

¹ Regulation (EU) 2023/857 of the European Parliament and of the Council of 19 April 2023 amending Regulation (EU) 2018/842 on binding annual greenhouse gas emission reductions by Member States from 2021 to 2030 contributing to climate action to meet commitments under the Paris Agreement, as well as Regulation (EU) 2018/1999

² Directive (EU) 2023/959 of the European Parliament and of the Council of 10 May 2023 amending Directive 2003/87/EU establishing a system for greenhouse gas emission allowance trading within the Union and Decision (EU) 2015/1814 concerning the establishment and operation of a market stability reserve for the Union greenhouse gas emission trading scheme

- Promoting research and development activities related to the production of sustainable aviation fuels;
- Organise awareness raising campaigns, capacity building of stakeholders with regard to the development of sustainable mobility.

The priorities set for the **Transport Connectivity Programme 2021-2027** are:

- Priority 1 "Development of railway infrastructure along the "core" and "comprehensive" trans-European transport network;
- Priority 2 'Road infrastructure development' on the core 'Trans-European Transport Network' and road connections;
- Priority 3 'Improving intermodality, innovation, modernised traffic management systems, improving transport security and safety';
- Priority 4 "Intermodality in urban environments".

They contribute to the implementation of the Green Deal, the EC Sustainable and Smart Mobility Strategy and the implementation of national transport policy.

The investments envisaged under the priorities of the Programme promote the use of environmentally friendly modes of transport and alternative fuels, improve the quality of road and rail infrastructure, promote intermodality and intelligent transport systems and thus contribute to reducing the harmful environmental impact of transport.

The envisaged investments under priority 1 will contribute to attracting passenger and freight traffic to rail by improving the quality of rail infrastructure.

Investments for the development of railway infrastructure are mainly concentrated along the Orient/East-Mediterranean corridor section running horizontally through the middle of the country.

Investments under Priority 3 will contribute to the development and extension of inland and maritime public transport ports for multimodal operations, modernisation and development of terminals and port facilities for combined transport, as well as to the development of the Gorna Oryahovitsa, Ruse and Varna railway nodes. Further investment is foreseen for the construction of alternative fuels infrastructure along the main strands of the Republican Road Network (RMP). Interventions are for road sections between some of the country's largest cities where transport is identified as an air pollutant. In addition, the construction of alternative fuels recharging infrastructure also in public transport ports will be supported.

Priority 3 sets out two main procedures as follows:

- **Intermodal Operators procedure, under** which a grant scheme with an intensity of up to 50 % is planned to support all intermodal operators for:
 - Purchase of equipment;
 - Construction/rehabilitation of railway/road infrastructure;

- Cargo handling sites;
- Deployment of IT systems and charging stations.

The investments will contribute to the development and expansion of intermodal combined transport terminals, thus creating the necessary conditions and preconditions for multimodal operations.

- **Alternative fuels procedure**, which will finance the construction of alternative fuels infrastructure under the LFS (TEN-T) as well as in TEN-T public transport ports (maritime and inland waterways).

The procedure aims at setting up a national scheme to support the construction of recharging infrastructure to enable the financing of the development of charging infrastructure for light-duty and heavy-duty electric vehicles, as well as in public transport ports/sea and inland waterway/TEN-T, to be launched in 2024.

Integrated Transport Strategy for the period up to 2030

The strategy outlines the main guidelines for the development of the national transport system in the period up to 2030.

The document sets out 3 strategic objectives, covering 9 strategic priorities, each containing a framework of specific objectives (targets). On this basis, measures have been identified that are most appropriate to achieve the relevant objectives.

The strategic objectives of transport policy until 2030 are to:

- Increasing the efficiency and competitiveness of the transport sector;
- Improving transport connectivity and accessibility (internal and external);
- Limiting the negative effects of the development of the transport sector.

The strategic priorities for transport development are:

- Efficient maintenance, modernisation and development of transport infrastructure;
- Improving the management of the transport system;
- Development of intermodal transport;
- Improving the conditions for implementing the principles of liberalisation of the transport market;
- Reducing fuel consumption and increasing the energy efficiency of transport;
- Improving the connectivity of the Bulgarian transport system to the Single European

Transport Area;

- Ensuring quality and affordable transport in all areas of the country;
- Limiting the negative impact of transport on the environment and human health;

- Enhancing the security and safety of the transport system.

A National Transport Model has also been prepared within the scope of the Strategy Paper, which has been developed for passenger and freight transport and is applicable to individual modes of transport within the country, international transport and transit.

The main measures affecting greenhouse gas emissions in the transport sector are:

T1 – Increasing share of public electric transport – rail, trolleybus, tramway, metro

Scope: Increasing the share of public electric transport. Increasing the share of electric rail transport – improving infrastructure; Increasing the share of electric rail transport – renewal of rolling stock; Increasing the share of electric mass public transport – infrastructure improvements; Increasing the share of electric mass urban transport – vehicle renewal

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2014 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: **WEM**

T2 – Increasing the share of biofuels

Scope: Biofuels are fuels produced from biomass and used in transport. They diversify the energy balance and reduce dependence on mineral fuels. The main types of biofuels are bioethanol, biodiesel, biogas, synthetic biofuels, biohydrogen, pure vegetable oils. In Bulgaria, ethanol and biodiesel production projects are the most promising. In the Energy from Renewable Sources Act (Article 47. (1)) steps are in place to achieve certain percentages of biodiesel and bioethanol in the respective fuels, as well as biofuel type requirements and sustainability criteria to be met. Biofuels will continue to be used in the coming years and the share of new biofuels will be gradually increased.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2012 – 2030

Sources of funding: State and municipal budget

Scenario in which the measure is included: WEM

T3 – Introduction of Intelligent Transport Systems on the national and urban road network

Scope: Intelligent Transport Systems (ITS) cover a wide range of technical solutions designed to improve transport, improving mobility and improving road safety. Telematics (a combination of telecommunications and informatics) uses advanced technologies to meet transport needs. Intelligent transport systems and telematic solutions help improve road safety, promote the efficiency of existing infrastructure used and contribute to reducing environmental pollution by controlling traffic and managing traffic volumes.

Urban intelligent transport systems may include integrated management of public transport charges, enhanced customer relationship management, traffic forecasts, improved traffic management, passenger information and road charging. These systems apply modern technologies to collect more and better data to perform accurate analysis and connect them through more efficient networks. The result is more efficient and more citizen-oriented transport on urban traffic. European funds with public and municipal cofinancing and, in some cases, the climate and private investment programme are the source of funding.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2014 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

T4 – Development and construction of intermodal combined transport terminals

Scope: The aim of the measure is to achieve a bilateral effect consisting, on the one hand, of increasing the usability of more environmentally friendly modes of transport and, on the other hand, of creating the right conditions to increase the added value of transport activity, with a general reduction in transport costs per unit of GDP.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2014 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

T5 – Promoting the use of hybrid and electric vehicles

Scope: In the framework of the Innovation Strategy for Smart Specialisation (ISS) 20212027, in the Thematic Area Clean Technologies, Circular and Low-Carbon Economy, one of the priority areas is the development and deployment of sustainable mobility technologies (batteries and hydrogen) based on hydrogen and other alternative fuels, connected infrastructure and eco-mobility.

One of the centres designated as a specific beneficiary is the HITMOBILL Competence Centre – Technology and systems for the generation, storage and consumption of clean energy. The Centre focuses on research, experimental development and knowledge transfer in the field of “Clean Energy generation, storage and consumption technologies and systems”.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2014 – 2030

Sources of funding: Climate Investment Programme, Private Investment

Scenario in which the measure is included: WEM

T6 – Rehabilitation and modernisation of existing road infrastructure to ensure optimal speed and optimal driving modes for motor engines

Scope: Rehabilitation and modernisation of existing road infrastructure to ensure optimal speed and optimal driving modes for motor engines.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2014 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

T7 – Reducing the relative share of private motor vehicle travel by improving and developing urban public transport and by developing non-motorised transport

Scope: Improving urban public transport and the development of non-motorised transport.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2012 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

T8 – Developing and stimulating cycling

Scope: Project-oriented approach – concrete implementation – 1. Design and implementation of new infrastructure for cycling, 2. Development of systems for the use of municipal bicycles; 3. Training and campaigns.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

T9 – Fiscal policy to stimulate the economy and limit the consumption of conventional fuels

Scope: Financial policy through tax incentives for producers and users of electric vehicles and better application of the 'polluter pays' and 'user pays' principles.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2014 – 2030

Sources of funding: Tax reductions and charges laid down in the State and municipal budgets

Scenario in which the measure is included: WEM

New (planned) measures:

T10 – Establishment of a national scheme to support the construction of recharging infrastructure

Scope: Alternative fuels procedure, which will finance the construction of alternative fuels infrastructure under the LFS (TEN-T) as well as in TEN-T public transport ports (maritime and inland waterways).

The procedure aims at setting up a national scheme to support the construction of recharging infrastructure to enable the financing of the development of charging infrastructure for light-duty and heavy-duty electric vehicles, as well as in public transport ports/sea and inland waterway/TEN-T, to be launched in 2024.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2024 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

T11 – Promoting the development of combined transport

Scope: The Ministry of Transport and Communications developed a National Plan for the Development of Combined Transport in the Republic of Bulgaria by 2030 (approved by Decision No 504 of the Council of Ministers of 21.7.2022). It is a basic document for the implementation of a policy to support sustainable modes of transport. An implementation programme linked to the 2030 time horizon and potential funding sources has also been proposed. Three groups of measures have been identified: (1) organisational and administrative, (2) operational and service support and (3) infrastructure. Their implementation is very important because it will increase the efficiency of the transport system by exploiting the advantages of combined transport (lower emissions of harmful substances, increased road safety, noise reduction due to reduced road transport, reduced use of conventional fuels in transport) over the use of road haulage only.

The plan will support projects to develop a network of modern intermodal terminals and identify incentives for business to implement more efficient and environmentally friendly transport solutions and chains. By 2030, the construction of intermodal terminals in Sofia and northern Bulgaria was planned, as well as a study on the need to build a terminal in Vidin. The document includes measures to develop logistics centres in Bulgaria and to improve the infrastructure connecting to existing port and rail-road terminals.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2024 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

T12 – Promoting the production and consumption of alternative renewable fuels, namely hydrogen, including by developing alternative fuels infrastructure

Scope: Developed and adopted in 2023, the National Roadmap to improve the conditions for unleashing the development potential of hydrogen technologies and hydrogen production and supply mechanisms sets the path for building a hydrogen industry. The aim of the Roadmap is to provide the basis for a coherent framework for the efficient, smooth and consistent uptake of hydrogen production, transport and use technologies in industry, energy, transport and household, to foster innovation and investment. The introduction of hydrogen electric transport is foreseen to start from urban bus transport under the responsibility of the municipalities. This approach is more cost-effective due to the possibility of scaling up with a large number of means of transport and recharging infrastructure with high regulated usability rates.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2025 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

T13 – National Recovery and Resilience Plan

Scope: Reforms envisaged: 1 – for sustainable urban mobility, 2- Green mobility – a pilot scheme to support sustainable urban mobility through measures to develop green, safe, functional and energy-efficient transport systems, 3- Ensuring effective access to integrated public transport, 4 – Electric mobility (Law on the promotion of electromobility).

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2024 – 2026

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

Measures with indirect effect on GHG emission reductions

T14 – Facilitate the informed choice of a transport vehicle to increase the number of vehicles purchased with lower emission levels (corresponding to Euro IV, V or VI).

T15 – Restructuring of urban streets;

T16 – Promoting sustainable driving;

T17 – Legislative provisions have been introduced to allow the establishment of LowEmission Zones (LEZ) in cases where the type and extent of ambient air pollution significantly increases the risk to human health and/or the environment or failure to meet the standards for harmful substances (pollutants) in ambient air and the deposition of

harmful substances (pollutants), approved by regulations of the Minister for the Environment and Water and the Minister for Health.

Adaptation measures from the Action Plan to the National Adaptation Strategy contributing to decarbonisation:

T18 – Development and implementation of a programme to strengthen the resilience of the road network to extreme weather events;

T19 – Development and implementation of a programme to strengthen the resilience of the rail network to extreme climatic events;

T20 – Regular update of the standards for the design of roads and railways.

Industry sector

The main measures to reduce greenhouse gases in the industrial sector are:

IP1 – Increasing energy efficiency in industry and reducing heat losses Scope: Improving efficiency in industrial final consumption sectors.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

IF2 – Increasing the use of natural gas in industry through new gas infrastructure

Scope: Improved transport infrastructure

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2013 – 2020

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

IY3 – Use of alternative fuels including the inclusion of renewable, bio and low-emission fuels in the energy mix of the gas grid Scope: Improved transport infrastructure

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

IF4 – Introduction of incentives to encourage the private sector to invest in R & D and innovation of widely used production methods aimed at optimum resource efficiency Scope: Improving efficiency in industrial final consumption sectors.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

IF5 – Promoting the exchange of good practices between businesses on raw material efficiency in manufacturing

Scope: Improving efficiency in industrial final consumption sectors.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

II6 – Industrial energy management systems Scope:

Reduction of greenhouse gases.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

IK7 – Energy efficiency audits and implementation of prescribed measures Scope:

Improving efficiency in industrial final consumption sectors.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

II8 – Prohibitions, restrictions and obligations under Regulation (EU) No 517/2014 on fluorinated greenhouse gases and Directive 2006/40/EC on emissions from air-conditioning systems in motor vehicles

Scope: Reducing emissions of fluorinated gases and replacing fluorinated gases by other substances. Regulation (EU) No 517/2014 on fluorinated gases (which entered into force on 1 January 2015) establishes a timetable for the gradual reduction of fluorinated gases by 2030, to be implemented by implementing a system of authorised units and prohibitions/restrictions.

The most important measures to reduce fluorinated greenhouse gas emissions in Regulation (EU) No 517/2014 are:

- Bans on the placing on the market of certain new devices;
- A prohibition on servicing F-gases with a global warming potential of 2500 or more;
- Requirement to separate gases from decommissioned equipment;

- Obligation to certify gas companies.

Greenhouse gases directly concerned: HFCS

Execution period: 2015 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

New measures to reduce greenhouse gas emissions in industry in the plan:

IY9 – Developing incentives for enterprises pursuing industrial symbiosis such as:

- use of by-products and/or waste products from one plant for the raw material of another;
- sharing utilities – energy, water or wastewater treatment to reduce costs and improve resource productivity and environmental performance;
- sharing services such as logistics, joint marketing (e.g. shared call centres) and counselling.

Scope: Improving efficiency in industrial final consumption sectors.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2021 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

RI10 – Promoting through additional financial compensation the purchase of net-zero technology final products with a high contribution to economic resilience

Scope: Improving efficiency in industrial final consumption sectors.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2021 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

RI11 – Provision of improved permitting procedures for the following strategic net zero technologies to enhance the interest of the investor:

- solar photovoltaic and solar thermal technologies;
- terrestrial and offshore renewable technologies;
- batteries and storage technologies;
- heat pumps and geothermal energy;
- electrolyzers and fuel cells;
- use of hydrogen as raw material and/or process agent;
- sustainably sourced biogas and biomethane;
- carbon dioxide capture, storage and recovery;
- network technologies.

Scope: Improving efficiency in industrial final consumption sectors.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2021 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

RI12 – Improving conditions for investments in net-zero technologies by:

- ensure equal access to all European and national financing opportunities (decarbonisation, modernisation and innovation fund, NRRP, etc.), including through State aid authorisation regimes for the decarbonisation of industry. Support for the provision of grants for decarbonisation and energy efficiency improvement projects for the energy intensive industry, including large companies participating in the Emissions Trading System;

- provide priority support for the deployment of new technologies with a low and medium technology readiness level (TRL) to decarbonise hard-to-decarbonise processes, including process emissions (CCSU, hydrogen, etc.);

- improving awareness among stakeholders;

- reducing the administrative burden to create projects in this area; –

- simplifying and streamlining permitting processes.

Scope: Improving efficiency in industrial final consumption sectors.

Greenhouse gases directly concerned: CO₂, N₂O

Execution period: 2021 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

RI13 – Promote the production and consumption of alternative renewable fuels, namely hydrogen, including by developing alternative fuels infrastructure

Scope: Developed and adopted in 2023, the National Roadmap to improve the conditions for unleashing the development potential of hydrogen technologies and hydrogen production and supply mechanisms sets the path for building a hydrogen industry. The aim of the Roadmap is to provide the basis for a coherent framework for the efficient, smooth and consistent uptake of hydrogen production, transport and use technologies in industry, energy, transport and household, to foster innovation and investment. The introduction of hydrogen electric transport is foreseen to start from urban bus transport under the responsibility of the municipalities. This approach is more cost-effective due to the possibility of scaling up with a large number of means of transport and recharging infrastructure with high regulated usability rates.

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2025 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

IY14 – Promotion of carbon capture and storage projects

Scope: Support for innovative low-CO₂ technologies in energy-intensive industries

Greenhouse gases directly concerned: CO₂, CH₄, N₂O

Execution period: 2028 – 2050

Sources of funding: Innovation Fund

Scenario in which the measure is included: WAM

IM15 – Ensuring access to decarbonised energy, infrastructure and raw materials

IY16 – Creating markets for low-carbon, circular products

IF17 – Increasing the use of natural gas in industry through new gas infrastructure

IP18 – Promoting the production of low-emission ammonia and its use as a raw material and energy carrier

IY19 – Introduction of an unobscured DAM ensuring a level playing field for all producers in relation to the pricing of greenhouse gas emissions

RI20 – Prioritising the application of circular economy principles, promoting the production of critical raw materials and deploying technologies to reduce scope 1 and scope 2 greenhouse gas emissions, including in hard-to-decarbonise processes

IY21 – Adaptation of legislation to promote the use of nuclear energy for the production of heat and electricity, hydrogen and other industrial applications, respecting the principles of ensuring nuclear safety, nuclear security and radiation protection and implementing good practices in radioactive waste management and spent nuclear fuel management.

In addition, public authorities should take into account sustainability criteria for zero-net technologies in public procurement or tenders.

In addition to the EU Emissions Trading System, European legislation on industrial emissions (integrated pollution prevention and control), reducing fluorinated greenhouse gases and controlling ozone depleting substances also contribute to reducing GHG emissions as well as emissions of harmful substances into the air.

Agriculture sector

The development of the agricultural sector and the implementation of the various measures are mainly governed by the Strategic Plan for Agriculture and Rural Development of the Republic of Bulgaria for 2023-2027 and the implementation of the sustainable agriculture component of the National Recovery and Resilience Plan of the Republic of Bulgaria.

The Strategic Plan for Agriculture and Rural Development 2023-2027 sets out 9 specific objectives (CSRs), including three objectives with a direct and indirect effect on climate and climate change:

JU 4 – Contribution to climate change mitigation and adaptation, including by reducing greenhouse gas emissions and enhancing carbon sequestration, as well as promoting sustainable energy;

OP 5 – Promoting sustainable development and efficient management of natural resources such as water, soil and air, including by reducing chemical dependency;

JR 6 – Contributing to halting and reversing biodiversity loss, improving ecosystem services and preserving habitats and landscapes.

To achieve the specific objectives, interventions are planned to support farmers in making voluntary commitments in farm management related to adaptation and mitigation of climate change and the protection and restoration of biodiversity and agricultural ecosystems.

Farmers are obliged to comply with basic standards in the areas of environment, climate change, public health, plant health and animal welfare. The basic standards cover certain statutory management requirements (SMRs) and standards for good agricultural and environmental condition of land (GAEC standards). These basic standards take greater account of environmental and climate challenges and the new environmental architecture of the Common Agricultural Policy, thus demonstrating higher environmental and climate ambition.

The environmental schemes for the climate, environment and animal welfare included in the Strategic Plan for Agriculture and Rural Development 2023-2027 contribute to achieving the objectives and priorities of the decarbonisation dimension with direct or indirect effects.

The following eco-schemes are designed to respond to the emission reduction needs of the agricultural sector and the protection and restoration of biodiversity and soil and are accessible for implementation by farmers: I.C.1 – Organic scheme (livestock);

I.C.2 – An scheme for maintaining and enhancing biodiversity and ecological infrastructure;

I.C.3 – Eco scheme to preserve and restore soil potential – promotion of green fertilisation and organic fertilisation;

I.C.4 – Eco pesticide use reduction scheme;

I.C.5 – Environmental maintenance scheme for permanent crops;

I.C.6 – An extensive maintenance scheme for permanent grassland;

I.C.7 – An scheme for maintaining and enhancing biodiversity in forest ecosystems;

I.C.8 – Eco crop diversification scheme.

To enhance farmers' knowledge and skills, the National Agricultural Advisory Service (NSPA) provides advice and individual advice to farmers on measures with an indirect and direct effect on greenhouse gas emission reductions for the following measures:

- Humus conservation activities (fertilisation – precision fertilisation, green fertilisation; liming; respectful tillage, crop rotation, anti-erosion activities, etc.);
- Water-saving and energy-saving irrigation technologies;
- Extensive grazing of the animals;
- Opportunities to exploit plant residues and threats from burning stubble.

Consultations on measures with a direct effect on greenhouse gas emission reductions shall cover the following areas:

- Improving manure storage and application;
- Low carbon manure processing practices (composting, anaerobic biogas processing, etc.).

The State Fund for Agriculture – Paying Agency (DFZ-PA) is a control body and makes payments on applications submitted by farmers applying for support. The intended use of the declared areas and their maintenance in good agricultural and environmental condition is established by checks by the Technical Inspectorate of the DFZ-PA. Non-compliance with the standards established by a check shall be penalised by reducing the payments to the farmer.

The measures in the Third National Climate Action Plan with a 2030 horizon and the National Air Pollution Control Programme 2020-2030 aim at reducing emissions from the sector's main sources. One of the main challenges faced by the CAP is to address the increasingly deteriorating agricultural production conditions due to climate change and the need for farmers to reduce their share of greenhouse gases, play an active role in mitigating climate change and providing renewable energy.

Based on the analysis of the main emission sources in agriculture, the following two main objectives are defined:

- Reducing and/or optimising greenhouse gas emissions from the agricultural sector;
- Raising awareness and knowledge of both farmers and administrations with regard to their actions and impact on climate change. These main objectives are the following priorities:
 - Reducing greenhouse gas emissions from agricultural land;
 - Reducing methane emissions from organic fermentation in livestock farming;
 - Improving manure management;
 - Optimising the use of plant residues in agriculture;
 - Improving rice field management and rice production technologies;
 - Improving farmers' and administration's knowledge of reducing emissions from the agricultural sector.

The measures foreseen in the Third National Climate Action Plan, with a horizon up to 2030 and the National Air Pollution Control Programme 2020-2030, which continue to contribute to reducing greenhouse gas emissions are:

A1 – Stimulating the use of appropriate crop rotations, especially with nitrogen-fixing crops

Scope: Crop rotation means the scientifically justified sequential rotation of crops at a time and place on a given arable area. The period required for all crops to pass through all fields in the order defined by the rotation scheme is referred to as a rotation period or rotation. The introduction of sustainable crop rotation that includes vegetation cover in winter and leguminous crops (legumes, soya, lucerne, clover) will protect soils from erosion and maintain organic carbon sequestration, which is a potential means of reducing greenhouse gas emissions.

Greenhouse gases directly concerned: CH₄

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

A2 – Management of degraded agricultural land through biological recultivation with herbaceous species typical of the region and implementation of erosion control measures and soil tillage methods

Scope: Soil erosion is a process of mechanical demolition and spreading of the soil mass from the action of water and wind. Under its influence, the amount of nutrients and humus in the soil is gradually reduced. As a result of erosion, the soil structure and water/air regime deteriorate. The combination of the specific natural and economic conditions in Bulgaria creates a high risk of developing soil degradation processes used in agriculture. The most common soil degradation processes include: water and wind erosion, pollution, reduction of organic matter (humus) stocks, compaction, acidification, salinisation, loss of biodiversity.

Greenhouse gases directly concerned: CH₄, CO₂, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

A3 – Introduction of irrigation technologies and water and energy saving, promotion of extensive farming

Scope: Irrigation of agricultural land will play an increasingly important role in the parallel impact of the development of the agricultural sector and the impact of climate change. Efficient and rational use of water is of the utmost importance both for soil health and for reducing the need to use additional energy when watering.

Greenhouse gases directly concerned: CH₄, CO₂, N₂O

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

A4 – Improving manure management and use

Scope: Manure production, treatment and management is one of the most significant sources of CH₄ greenhouse gas emissions in agriculture. All activities aimed at manure storage and treatment must take into account both the type of manure, solid or liquid, as well as the collection and treatment technologies.

Investment support is essential to motivate farmers to build such expensive facilities.

Greenhouse gases directly concerned: CH₄

Execution period: 2013 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

A5 – Introduction of low carbon manure handling practices, e.g. composting, conversion of manure to biogas under anaerobic conditions

Scope: By introducing low-carbon practices for manure processing, emissions from manure storage can be reduced. This requires considerable knowledge and experience at regional level, as the effectiveness of the implementation of the measure depends on the conditions under which it is applied. For this reason, it is advisable to build model farms in the various production areas of the country in order to gain practical experience that can be presented to farmers.

Given the resource intensity of such an investment and the need to change the production process, investment support is also advisable.

Investment support is essential to motivate farmers to build such expensive facilities.

Greenhouse gases directly concerned: CH₄

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Execution period: 2014 – 2030

Scenario in which the measure is included: WEM

A6 – Improving farmers' awareness and knowledge about the possible use of plant residues and the threats posed by burning stubble

Scope: The use of plant waste in agriculture requires both a change in or adaptation of the production process and investment in new machinery and machinery. All of this requires a significant financial resource, which is why it is appropriate to support them.

Effective recovery of waste will reduce the need for burning the stubble.

Greenhouse gases directly concerned: CH₄

Execution period: 2014 – 2030

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WEM

New measures to reduce greenhouse gas emissions in the plan:

A7 – Application of the Good Agricultural Practice Rules for the control of ammonia emissions into the air from agricultural sources on the basis of the UNECE Framework Code on Good Agricultural Practice for Reducing Ammonia Emissions: good practices for lowemission fertiliser/manure application, good manure management practices and strengthening the implementation of the Nitrates Directive).

Scope: Promoting the introduction and continued use of environmentally friendly nitrogen management methods in agriculture; Preventing and reducing nitrogen pollution of water from agricultural production; better management of organic fertilisers; reducing the use of fertilisers/manure on arable land; improved livestock management; actions to improve the management of arable land.

Greenhouse gases directly concerned: CH₄, CO₂, N₂O

Execution period: 2023 – 2027

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

A8 – Promotion of organic farming

Scope: The objective of the measure is to support and enhance the competitiveness of organic farming, to increase biodiversity and landscape diversity, and to maintain and improve soil fertility and water quality. The measure will contribute to reducing greenhouse gas emissions by using organic fertilisers instead of mineral fertilisers.

Greenhouse gases directly concerned: CH₄, CO₂, N₂O

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Execution period: 2023 – 2027

Scenario in which the measure is included: WAM

A9 – The Strategic Plan for Agriculture and Rural Development 2023-2027 – Eco Crop Diversification Scheme

Scope: — The requirements of the intervention provide for a higher number of crops grown on the holding. The increased number of different crops contributes to minimising risks to the farm due to climate change (adaptation and mitigation) through crop diversification. The intervention aims to improve crop rotation, which increases the humus and nutrient content of the soil and protects against erosion by ensuring the presence of permanent soil cover to avoid soil drainage and compaction. *Greenhouse gases directly concerned: CH₄, CO₂, N₂O*

Execution period: 2023 – 2027

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

A10 – Strategic Plan for Agriculture and Rural Development 2023-2027 – A scheme to preserve and restore soil potential.

Scope: The intervention provides for two main beliefs of the introduction of external organic matter into the soil: cultivation of suitable types of intermediate non-production crops with subsequent green fertilisation and introduction of external organic matter subject to circular economy products (organic matter treated from waste biomass). The intervention aims to reduce the use of artificial fertilisers, replacing them with natural nutrients formed as a result of the introduction of external natural organic matter.

Greenhouse gases directly concerned: CH₄, CO₂, N₂O

Execution period: 2023 – 2027

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

A11 – Strategic Plan for Agriculture and Rural Development 2023-2027 – Eco scheme for extensive maintenance of permanent grassland

Scope: The intervention will prevent the loss of the pastoral ecosystem by reducing the processes of sinking and erosion, ruiralisation and the development of atypical species, which significantly improves the capacity of grassland systems to absorb carbon and mitigate climate change.

Greenhouse gases directly concerned: CH₄, CO₂, N₂O

Execution period: 2023 – 2027

Sources of funding: European funds with state and municipal co-financing, state and municipal budgets

Scenario in which the measure is included: WAM

Adaptation measures from the Action Plan to the National Adaptation Strategy contributing to decarbonisation:

AA1 – Development of suitable irrigation systems

AA2 – Maintenance of existing pasture

AA3 – Improving soil structure maintenance

AA4 – Development of insurance and risk management programmes

Waste management sector

Waste prevention is the most efficient way to improve resource efficiency and to reduce the environmental impact of waste.

The Waste Framework Directive 2008/98/EC introduces a waste management hierarchy for institutions, businesses and households to tackle waste and prioritises measures in the following order:

- Waste prevention;
- Preparing for re-use;
- Recycling;
- Other recovery, e.g. energy recovery;
- Disposal (controlled disposal, incineration without energy recovery, etc.).

The measures of the Third National Climate Action Plan 2013-2020 to be continued and upgraded until 2030 are:

Of 1 – Continuing and increasing separate collection of green waste in municipalities;

2 – biogas capture and incineration in all new and existing regional landfills;

Of 3 – Acquisition and incineration of biogas in recultivated municipal landfills;

Of 4 – Assessment of the energy potential of biogas from regional landfills that are planned to be re-cultivated;

Of5 – Introduction of anaerobic stabilisation of sludge with biogas capture and combustion in new installations and installations undergoing reconstruction in settlements with more than 20 000 population equivalents;

Of 6 – Construction of municipal facilities for the recovery of biodegradable waste, with energy production and compost;

Of7 – Introduction of differentiated charges for waste generated.

New policies and measures in the Waste sector

National Waste Management Plan 2021-2028

Regulatory, environmental, economic measures; implementation: 2021 – 2028

Plays a key role in the effective and efficient management of waste in the country. The main objectives of the Plan include reducing the harmful impact of waste by preventing and promoting re-use, increasing the amount of waste recycled and recovered, and reducing the amount and risk of landfilled municipal and other waste.

Five programmes have been developed to achieve the objectives as part of the Plan, namely:

- A waste prevention programme with a sub-programme on food waste prevention;
- A programme to achieve the preparing for re-use and recycling targets for municipal waste;
- A programme to achieve the targets for recycling and recovery of construction and demolition waste;
- A programme for the achievement of recycling and recovery targets for mass waste with a sub-programme on packaging waste management;
- A programme to reduce the quantities and risk of landfilled municipal and other waste.

The programmes contain both investment measures and non-investment – soft measures. Investment measures mainly include the construction of infrastructure. Soft measures include a diverse range of activities, including regulatory changes; the appointment of officers; training of staff; preparation of calls for projects; preparation of projects; development of methodologies, instructions and other administrative acts; implementation of control activities; conducting information campaigns; development and implementation of IT systems, etc.

The estimated value of the programmes quantified under this approach is BGN 1 428.51 million.

Investment measures account for 78 % of planned expenditure in the 2021-2028 NRMP, while soft measures account for 22 % respectively. The Programme to reduce the quantities and risk of landfilled municipal waste accounts for 51 % of the total planned investments in the NMSP 2021-2028, followed by the Programme to reach the preparing for re-use and recycling targets for municipal waste by 27.4 % of the investments.

Measures of direct and indirect relevance to reduce greenhouse gas emissions:

Completion/upgrading of regional municipal waste management systems

Description: Programme for reaching preparing for re-use and recycling targets for municipal waste. Performance indicator: share (%) of implemented funds; number of contracts completed; number of installations built. Responsible institutions: The MA of the AOC, municipalities; RMWS

Greenhouse gases directly concerned: CH₄, N₂O

Execution period: 2021 – 2030

Sources of funding: Environment Programme 2021-2027

Construction of municipal sites for the free delivery of separately collected household waste, including large waste, and other separately collected waste in all settlements with a population of more than 10 000 inhabitants

Description: Programme for reaching preparing for re-use and recycling targets for municipal waste. Performance indicator: Number of sites built. Responsible institutions: The MA of the AOC, municipalities, RMAs.

Greenhouse gases directly concerned: CH₄, N₂O

Execution period: 2021 – 2030

Sources of funding: Environment programme 2021-2027, municipal budgets.

Making composters available to households free of charge for green and other bio-waste

Described: Waste prevention programme with a sub-programme on food waste prevention. Performance indicator: Number of composters provided to households. Responsible institutions: municipalities, households. *Greenhouse gases directly concerned: CH₄, N₂O*

Execution period: 2021 – 2030

Sources of funding: municipal budget, PEF 2021-2027, PUDOOS.

Reducing waste from paper and other office supplies by implementing national and sectoral e-government programming documents

Description: Waste prevention programme with a sub-programme on food waste prevention. Performance indicator: EGovernment projects implemented. Responsible institutions: MA of road 2021-2027, state and municipal administrations.

Greenhouse gases directly concerned: CH₄, N₂O

Execution period: 2021 – 2030

Sources of funding: Technical Assistance Programme 2021-2027.

Implementation of measures for PEs when issuing/updating integrated waste permits by the RIEW

Description: Waste prevention programme with a sub-programme on food waste prevention. Performance indicator: Number of permits in which a PO requirement is included. Responsible institutions: EEA, Business Organisations within the scope of the CoR.

Greenhouse gases directly concerned: CH₄, N₂O

Execution period: 2021 – 2030 Sources of funding: EEA.

Construction of waste recycling plants

Description: Programme for reaching preparing for re-use and recycling targets for municipal waste. Performance indicator: Number of waste recycling installations built. Responsible institutions: Ministry of the Environment and Water, municipalities, legal entities with a profit-making purpose.

Greenhouse gases directly concerned: CH₄, N₂O

Execution period: 2021 – 2030

Sources of funding: PUDOOS, Environment Programme 2021-2027, municipal budgets, private investment.

Separate collection and recycling of construction waste, including in industrial areas where proven necessary, Implementation Indicator: Number of projects implemented for separate collection and recycling of construction waste. Responsible institutions: Municipalities, legal entities with a profit-making purpose, the MA of the AOC.

Description: The programme to reach the targets for recycling and recovery of construction and demolition waste.

Greenhouse gases directly concerned: ML₂, CH₄, N₂O

Execution period: 2021 – 2030

Sources of funding: InvestEU; municipal budgets.

Energy sector

For the energy sector, most of the measures impacting decarbonisation are included in the sections on renewable energy, energy efficiency, internal market and energy security, as the overall changes in these dimensions lead to GHG emissions reductions.

All listed measures of the Third National Climate Action Plan (2013-2020) have been extended until 2030 as follows:

- Upgrade of cogeneration plants and central heating boilers with natural gas turbines;
- Reduction of losses from distribution and transmission networks;
- Reduction of losses in heat networks;
- Fuel switching from coal to natural gas;
- Increase of high-efficiency cogeneration;
- Increase of the share of energy for heating and cooling from renewable energy sources;
- Improving production efficiency in existing coal-fired power plants.

The main aggregate measures of direct and indirect relevance to reduce greenhouse gas emissions are:

EN1 – Promoting and facilitating the development of renewable self-consumption and establishing renewable energy communities

EN2 – Promoting the use of high-efficiency heating and cooling systems, the introduction of innovative geothermal and solar technologies and the use of waste heat and cold

EN3 – Promoting the use of biomass for centralised and local heat production, subject to the requirements of Article 28 (2) to (7) and (10) of Directive (EU) 2018/2001.

EN4 – Promoting the production of advanced biofuels, renewable liquid and gaseous transport fuels of non-biological origin, recycled carbon fuels and renewable electricity, for road and rail transport, and for air transport, sustainable aviation fuels

EN5 – Creating offshore wind energy development

EN6 – Introduction of a planning process for priority areas for the accelerated development of wind power plants

EN7 – Introduction of requirements for the use of energy from renewable sources in buildings

EN8 – Optimising the regulatory framework for applying the increased requirements of Directive (EU) 2018/2001 with regard to sustainability criteria and GHG emissions reduction

EN9 – Introduction of obligations on fuel and electricity suppliers to meet the target in the transport sector

EN10 – Creating conditions for the development and use of advanced biofuels, renewable liquid and gaseous transport fuels of non-biological origin and recycled carbon fuels

EN11 – National Recovery and Resilience Plan of the Republic of Bulgaria (NRRP) under component 2.B.1 Low-carbon economy:

Scope: Investment 6 (C4.I6): 'Support for new renewable electricity generation and electricity storage capacity'; Investment 2 (C4.I2) 'Support for renewable energy for households; Investment 5: (C4.I5) 'Pilot support scheme for the production of green hydrogen and biogas'; Investment 3: Energy-efficient municipal outdoor artificial lighting systems; Investment 1: Energy efficiency of building stock

EN12 – Introduction of a national financing mechanism for energy efficiency – National Decarbonisation Fund

EN13 – National programme for energy efficiency in multi-apartment buildings

EN14 – Long-term national strategy to support the renovation of the national stock of residential and non-residential buildings by 2050

EN15 – Establishment of a mandatory energy efficiency scheme (reduction of fuel and energy consumption in final energy consumption);

EN16 – Households affected by the mandatory withdrawal of stoves to heat with natural gas (reconnection and new connections), district heating (reconnection and new connections) or eco-design heating appliances), in accordance with the National Air Pollution Control Programme 2020-2030.

EN17 – Creating the transition from coal to low-carbon fuels by implementing the following projects:

It is appropriate to provide new gas transmission infrastructure for transmission to thermal power plants and other potential users in coal regions in order to create market conditions for the modernisation of thermal power plants and other energy consumers, to switch from coal to natural gas.

This will also create the necessary conditions for flexible and efficient operation of the installations after their modernisation, in line with the commitments to decarbonise the energy sector and transition to net carbon neutrality.

The use of fuels other than conventional fuels for primary energy production should be linked to the implementation of the following measures:

- Development and adoption of a national analysis of sustainable biomass potential from all sectors (including but not limited to forests and agriculture) and sustainability criteria, taking into account the sustainability criteria of Directive (EU) 2018/2001.

No similar national analysis has been carried out for the Gori sector. A similar analysis was developed in the National Forest Biomass Energy Action Plan 2018-2027.

- Synchronisation between strategic documents on forest management and use. When reviewing, updating and developing strategic documents, they must be consistent with each other and with the INECPs. Such strategic documents, which need to be revised, updated, aligned with the INECPs, may include: Forest Sector Report, Annex to the Action Plan of the National Strategy for Adaptation to Climate Change of the Republic of Bulgaria, National Strategy for the Development of the Forest Sector, National Action Plan for Forest Biomass Energy, National Renewable Energy Action Plan.

An evaluation was carried out on the implementation of the National Strategy for the Development of the Forestry Sector in the Republic of Bulgaria 2013-2020 (NSRGRB 2013-2020). In accordance with Article 9 (1) (1) of the Forests Act, a Strategic Plan for the Development of the Forestry Sector 2014-2023 (SDPFS 2014-2023) was drawn up, setting out the specific actions for the implementation of the strategic objectives, priorities and measures set out in the NSRGRB. In view of the expiry of the NSRGRB 2013-2020, a draft of the NFRGRB was prepared in 2022 and is due to be approved by the Council of Ministers in 2030. Monitoring and updating of the National Action Plan for Forest Biomass Energy, 2018-2027 is needed.

- Synchronisation between strategic documents related to the agricultural sector. When reviewing, updating and developing strategic documents, they must be consistent with each other and with the INECPs. The strategy papers, which will be reviewed, updated and aligned with the INECPs, may include plans for the implementation of the Common Agricultural Policy for the period after 2020;

- Synchronisation between strategy papers in relation to biodiversity conservation. When reviewing, updating and developing strategic documents, they must be consistent with each other and with the INECs. This includes the draft Biodiversity Strategy of the Republic of Bulgaria and the National Plan for the Conservation and Sustainable Use of Biological Diversity and Genetic Resources, to be approved by the Council of Ministers;

- A synchronisation of the INECs with the water management strategy papers should be ensured. When reviewing, updating and developing strategic documents, they must be consistent with each other and with the INECs. This includes the Flood Risk Management Plans (FRMPs) for the period 2022-2027, as well as the draft updated River Basin Management Plans (RBMPs) for the four river basin management areas. The activities set out in the Plan need to be consistent with the measures in the RBMPs and RBMPs in the four river basin management areas, throughout Bulgaria, and not only in Natura 2000 areas;

- Synchronisation between waste sector strategy documents.

The implementation of the above additional measures is considered to have a positive impact on both primary energy production and LULUCF sectors.

Adaptation measures from the Action Plan to the National Adaptation Strategy contributing to decarbonisation:

ENA1 – Inventarisation of strategies, policies, plans, standards, site selection requirements, energy infrastructure design norms, etc. to identify those in which climate sustainability considerations should be included;

ENA2 – Mainstreaming climate resilience considerations in the energy sector's investment plans by identifying climate risks in terms of their likelihood and impact;

ENA3 – Including seasonal and long-term projections of climate change in seasonal and long-term forecasts of electricity demand;

ENA4 – Mainstreaming climate resilience into water resource management and related decisions affecting the operation of large hydropower plants;

ENA5 – Mainstreaming climate resilience in the design of new power plants;

ENA6 – Development of maps showing climate risk areas associated with transmission and distribution infrastructure to inform decisions on which parts of the pyr networks require climate resilience actions;

ENA7 – Continue monitoring of causes of misses of the pyr system and classify climate/weather causes in order to understand which climate risks lead to most interruptions and to identify trends in their frequency;

ENA8 – Developing a cost-benefit analysis to assess whether additional sections of the distribution system need to be replaced by underground cables, taking into account changes in frequency and severity of extreme events and consequent network damage.

Land use, land use change and forestry sector (LULUCF)

Forests are a major carbon dioxide (CO₂) sink and are at the forefront of carbon sequestration through photosynthesis. They are an important link in the global carbon cycle due to their ability to capture CO₂ from the atmosphere and retain it in their biomass, dead forest pools/MNE (and forest soil). Tree species growth is largely a net carbon accumulation, so assessing and forecasting forest condition and productivity is essential to analyse the evolution of carbon emissions. In addition, the growth of woody biomass in forests plays a role in reducing greenhouse gas concentrations in the atmosphere. For these reasons, an analysis of the condition of forest ecosystems and forest resource management methods is of interest in view of the possibility of increasing the potential of forests as sinks.

Bulgaria has a significant forest resource and its sustainable management and development is an important factor in reducing greenhouse gases. The country's forest areas account for one third of its area.

In the long term, it is necessary to further develop the idea of sustainable and multifunctional forest management aimed at maintaining or increasing timber stocks and thus carbon in forests, while ensuring sustainable annual harvesting of wood and wood products.

Forests Act

Forest planning is carried out at three levels and includes a National Strategy for the Development of the Forestry Sector in the Republic of Bulgaria and a Strategic Forest Development Plan, Regional Forest Development Plans and Forestry Plans and Programmes.

Forest plans and programmes shall determine the eligible amount of use of forest resources and the guidelines for achieving forest management objectives for a period of 10 years. The Forestry Act does not allow for the reduction of the existing looseness in the territory of the Republic of Bulgaria and in the territory of municipalities whose looseness is less than 10 % of the total percentage of forest land in Bulgaria. Land use change in forest areas is only possible in certain specific cases.

National Strategy for the Development of the Forestry Sector in the Republic of Bulgaria 2013-2020

The implementation of the National Strategy for the Development of the Forestry Sector in the Republic of Bulgaria 2013-2020 was finally monitored by an inter-ministerial working group with the participation of the non-governmental sector.

The National Strategy for the Development of the Forestry Sector in the Republic of Bulgaria for the period up to 2030 is to be approved by the Council of Ministers.

A new Strategic Plan for the Development of the Forest Sector will be drawn up on the basis of the approved document.

The process also includes the final monitoring of the implementation of the Strategic Plan for the Development of the Forest Sector 2014-2023.

Strategic Plan for the Development of the Forest Sector 2014-2023 (SPDP)

This plan was developed with the financial support of the European Social Fund under the project 'Strategic planning in Bulgarian forests – guarantor of effective management and sustainable development' under the Operational Programme 'Administrative Capacity'.

The implementation of the operational objectives with the corresponding budget, timetable, expected results, performance indicators, responsible institutions and stakeholders is regulated in specific sub-activities in the SDPSG as follows:

Operational objective 1: 'Increase of forest area, tree stocks and carbon stocks in forest areas';

Operational objective 2: "Improving forest management and use";

Operational objective 3: "Increasing the effectiveness of preventing and combating forest fires and illegal activities in forests";

Operational objective 4: "Increasing the resilience and adaptability of forest ecosystems to climate change".

The achievement of these objectives is also ensured by the implementation of the Programme of Measures for the Adaptation of Forests in the Republic of Bulgaria and the mitigation of the negative impact of climate change on them.

The existing provisions of the Law on the ownership and use of agricultural land provide that the decision of the municipal council determines the rules for the use of meadows and pastures each year. Their scope shall include:

- *An action plan for grazing;*
 - Parts of grassland and pasture, mainly for mowing;
 - Measures for the conservation, maintenance and improvement of pastures, such as shrub and other undesirable plant species, anti-erosion activities, fertilisation, temporary fencing;
 - Parts of grassland and grassland for artificial grassland for planting with appropriate grass mixtures;
 - Use, prohibitions and restrictions depending on landscape, soil, climate and other natural conditions.

With regard to arable land, Article 7 of the Agricultural Land Ownership and Use Act provides that erodible, contaminated, saline, acidic and water-underground agricultural land is restored and improved on the basis of a range of activities or technologies that operate on the basis of pre-designed, coordinated and approved technologies and projects approved by the Council of Experts.

Valleys, quarries and other areas with impaired soil profile, ashes, tailings ponds, landfills and other landfills, old river beds, roads of abandoned canals, roads, railways and construction sites after dismantling engineering equipment, linings and superstructures are subject to recultivation. Recultivation is based on a pre-established, agreed and approved project, which is an integral part of the construction project. The procedure for using humus after removal, recultivation, land improvement and acceptance of recultivated areas are laid down in Regulation No 26 on land reclamation, improvement of low-productivity land, removal and utilisation of the humus layer.

To date, the report on the second mid-term monitoring and evaluation of the implementation of the Strategic Plan for the Development of the Forest Sector 2014-2023 has been validated.

One of the main strategic documents containing measures on land use, land use change and forestry is the third NCCP 2013-2020, whose measures will continue after 2020:

Measures contributing to the reduction of greenhouse gas emissions are:

LULUCF 1 – Use of 'unafforested areas for afforestation' in forest areas

Description: The realisation of the measure is important to achieve the objectives of the plan, as forests are a major carbon sink and a reservoir of 9095 % of the total carbon removals within the LULUCF sector. Increasing forest land plays an important role in offsetting greenhouse gas emissions in other sectors. With the uptake of unafforested areas for afforestation in forest areas in the long term, the capacity of forests as a greenhouse gas sink will also increase.

LULUCF 2 – Increase of area for urban and suburban parks and green areas

Description: The expansion of urbanised areas and intensive construction in recent years is a prerequisite for significant greenhouse gas emissions. Increasing the areas of urban and non-urban parks and green areas and maintaining them in good condition will contribute to increasing the absorption of greenhouse gases and improving the quality of the living environment. The implementation of the measure will also contribute to the gradual achievement of the rules laid down in the CAP for green areas.

LULUCF 3 – Restoration and sustainable management of wetlands. Protection and conservation of wetlands in forest areas, peatlands, marshes

Description: Wetlands are highly biodiverse and play an important role in carbon sequestration as they are among the most productive ecosystems. The restoration and conservation of wetlands in forest areas and their appropriate management will also increase their effectiveness as carbon pools. The NSPRZSR 2023-2027 provides for intervention for the construction or renovation of areas for widespread public use.

LULUCF 4 – Restoration and maintenance of protective forest belts and new anti-erosion afforestation

Description: In addition to the direct carbon absorption effect of afforestation in these belts, there is a significant indirect effect related to the prevention of wind erosion after reclamation.

No such measure has been implemented under the RDP 2014-2020. The NSPRZSR 20232027 provides for the implementation of intervention II.D.10 – Allocation and Recovery.

LULUCF 5 – Increasing density in listed natural and artificial plantations

Description: Activities to increase density in the listed plantations by supporting their natural regeneration or by other means.

LULUCF 6 – Introduction of a new measure/activity related to the creation of crops of fastgrowing tree species for the production of wood for energy purposes (short rotation plantations)

The Action Plan of the National Strategy on Adaptation to Climate Change provides for the following adaptation measures in the Forests Sector:

GA1 – Modelling the potential behaviour of the most important tree species at present and those species that may have potential in Bulgaria in a future changing climate, throughout the country and under different climate change scenarios and different timeframes and taking into account different topographical parameters

DA2 – Developing spatially linked models for the risk of natural disturbances from wind, fires, insects and diseases, etc.

GA3 – Strengthening the existing forest resource through enrichment and proactive management of risky plantations

GA4 – Restoration of forest areas severely affected by natural disturbances or drying and afforestation to improve forest protection functions with regard to water and soil
DA5 – Creation of short rotation plantations to produce biomass

GA6 – Conservation of highly biodiverse sites, plots of old-age forests and plots with available habitat (biotope) trees

Biodiversity and ecosystems

BEA1 – Development and adoption of the new Biodiversity Strategy and its Action Plan and a new Green Infrastructure Strategy for ecosystem-based management, conservation, restoration and CCA

BEA2 – operationalising ecosystem-based environmental impact monitoring and assessment

BEA3 – Creation of environmental carbon accounts BEA4

– Linking of carbon and environmental accounts

BEA5 – Regional/local red lines to prevent the loss of ecosystem services vital for ACIs

BEA6 – Restoring ecosystems – a long-term business opportunity